Multistage Relation Network With Dual-Metric for Few-Shot Hyperspectral Image Classification

Jun Zeng, Zhaohui Xue, *Member, IEEE*, Ling Zhang, Qiuping Lan, and Mengxue Zhang

***Abstract*— Recently, few-shot learning (FSL) has exhibited great potential in the hyperspectral image (HSI) classification due to its promising performance under a few training samples. Although existing FSL methods have achieved great success, some limitations can still be witnessed. On the one hand, current methods mainly rely on a single metric to identify, which cannot effectively represent the class distribution with few labeled samples. On the other hand, existing methods usually only use the last deep feature of the feature extractor, which may lead to the under-utilization of scarce labeled samples. To overcome the above issues, a novel multistage relation network with dual-metric (DM-MRN) is proposed for few-shot HSI classification. First, a sample recombination strategy is designed to increase the variety of classification tasks in the training period. Second, an embedding module is employed to extract deep features of the input image patches. Third, we propose two relation modules: image-to-class (I2C) block and image-to-image (I2I) block. The I2C block is designed to compute the I2C-level relation score between second-order features, and the I2I block is conceived to generate the I2I-level relation score between first-order features. Finally, DM-MRN is constructed by integrating one embedding module, two I2C blocks, and one I2I block. In addition, an adaptive weighting strategy is designed to fuse the obtained relation scores, and classification can be achieved by assigning each query sample to the class with the highest value of the fused relation score. Extensive experiments carried out on five popular HSI datasets demonstrate that the proposed method outperforms other traditional and advanced models under few training samples in terms of classification accuracy and generalization performance, i.e., the performance improvement in terms of OA is around 0.30%–27.98% under ten labeled samples per class.**

***Index Terms*— Classification, deep learning (DL), few-shot learning (FSL), hyperspectral image (HSI), multistage structure, relation network.**

I. Introduction

HYPERSPECTRAL image (HSI) can capture hundreds of narrow continuous spectral signals through the specific electromagnetic spectrum, making it possible to accurately identify ground objects [1], [2]. As a fundamental technique for various HSI applications, HSI classification has been widely used in many fields including precision agriculture, environmental monitoring, biotechnology, food analysis, military investigation, and mineral survey [2], [3], [4], [5], [6], [7], [8], [9].

In recent years, deep learning (DL) has been widely used in HSI classification [10], [11], [12], [13], [14]. In particular, 2-D convolutional neural networks (2-D-CNNs) [15], [16], [17], [18], [19] and 3-D convolutional neural networks (3-D-CNNs) [20], [21], [22], [23], [24], [25], with the powerful ability of modeling spatial–spectral information, are popular in HSI classification. In addition, the method based on transformer framework is also widely sought after by researchers due to its advantages in processing sequence data, and some advanced methods such as SpectralFormer [26], spectral–spatial feature tokenization transformer (SSFTT) [27], and local transformer with spatial partition restore network (SPRLT-Net) [28] have been derived one after another. However, the above methods are easily overfitting with few training samples, especially for 3-D-CNN which usually needs numerous training samples. At the same time, the labeled samples are very limited in real classification scenarios since the manual collection of labeled samples is time-consuming, laborious, and expensive. In this context, building a DL model for HSI classification with few labeled samples is advocated by many researchers [13].

An effective solution is graph convolutional networks (GCNs), which merely needs a few labeled samples to construct the spectral or spatial information between nodes and their neighbors in their training phase. Shahraki and Prasad [29] proposed an HSI classification model of cascading 1-D CNN and GCN, and designed a robust spatial–spectral semi-supervised adjacency matrix. Qin et al. [30] proposed a semi-supervised learning (SSL) model based on spectral–spatial GCNs, and extended the original GCN to the second-order version. Yang et al. [31] proposed a spectral–spatial graph using the inductive representation learning network, which constructed graphs by sampling and aggregating features from the local neighborhood of nodes. Gong et al. [32] introduced a superpixel spectral–spatial feature fusion graph convolution network, which can comprehensively use spectral, spatial, and spectral–spatial information. Considering the complexity of the adjacency matrix constructed by GCNs in memory and time is positively correlated with the number of pixels in HSI, Hong et al. [33] developed a

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minibatch GCN (miniGCN) for HSI classification, which can train a large-scale graph network in a minibatch fashion to reduce the operation cost of GCNs.

Recently, the combination of active learning (AL) and SSL can make use of both labeled and unlabeled samples, which provides another way to solve the insufficient label problem in HSI classification. Zhang et al. [34] introduced a strategy based on SSL to integrate spatial information and proposed an AL model that combined this strategy for HSI classification. Liu et al. [35] designed a super-pixel-guided semi-supervised AL method, which can actively select pseudolabeled samples with rich information. Yao et al. [36] designed semi-active CNNs, which integrate three kinds of AL schedules to reduce data distribution discrepancies between the training set and the testing set, thus having preferable generalization ability. However, it cannot be ignored that both AL and SSL need some manual intervention to decide the learning targets, which means that this method may cost more manpower, and there may be deviations and misleading caused by manual intervention.

At present, some typical FSL methods, including the siamese network, prototypical network, and relation network, have been increasingly exploited for HSI classification. These methods focus on transforming data such that they can be distinguished via some similarity metrics. Besides, they have the advantage of end-to-end training.

1) Siamese network, composed of two subnetworks sharing the same parameters, can determine whether a pair of input data are matched by similarity calculation. The early research, such as siamese CNN [37], is not an end-to-end solution for HSI classification. In order to remove the extra classifier, Rao et al. [38] proposed an end-to-end version of siamese CNN based on 1-D convolution. Subsequently, due to the limitation that 1-D convolution ignores the spatial information, Rao et al. [39] improved their previous research by proposing a siamese CNN with 3-D adaptive spatial–spectral pyramid pooling layer. To leverage unlabeled and labeled samples, Jia et al. [40] proposed a semisupervised siamese network that embeds siamese network into an SSL scheme. Later, considering that the combination of 1-D and 2-D convolution can significantly reduce the number of parameters compared with 3-D convolution when extracting the spectral–spatial features of HSI, Xue et al. [41] proposed a lightweight spectral–spatial siamese network for few-shot HSI classification.
2) Prototypical network performs classification by measuring the distance between the prototype and query sample. In order to well combine the spatial and spectral information, Tang et al. [42] designed a local pattern coding algorithm based on spatial neighborhood correlation, and further proposed a spatial–spectral prototypical network for HSI classification. Similarly, Xi et al. [43] proposed a deep prototypical network with hybrid residual attention (DPN-HRA) to enhance the critical spatial–spectral features while suppressing the useless features. In order to transfer the novel class samples to the embedding-feature space to generate the global prototypical representation of these samples, Zhang et al. [44] proposed a global prototypical network (GPN) for FSL. Later, in order to extract more representative features, Li et al. [45] combined semi-supervised clustering and AL with the prototypical network. To alleviate the overfitting issue in few-shot HSI classification, Wang et al. [46] proposed an adversarial prototypical network (APNet).
3) Relation network consists of two modules, the embedding module and the relation module. The former is used to extract deep features from input data, while the latter is used for classification by calculating the relation score between support features and query features via a neural network. Similar to the relation network proposed in [47], Deng and Shi [48] proposed a relation network for a few-shot HSI classification based on 2-D convolution. Subsequently, in order to explore spectral and spatial information simultaneously, Rao et al. [49] proposed a spatial–spectral relation network based on 3-D convolution. However, in reality, the imbalance of training sets samples often occurs. To solve this problem, Ma et al. [50] proposed a task-specific learning network for HSI classification.

Although current FSL methods are encouraging, they still have some limitations: 1) the single metric is difficult to effectively represent class distribution when training samples are scarce and 2) due to the scarcity of labeled samples, the deep features in different stages of feature extraction should be fully utilized.

In this article, we propose a novel multistage relation network with dual-metric (DM-MRN) for HSI classification. Compared with the existing methods, the important difference is that we simultaneously use two metrics to complete the few labeled samples classification of HSI. First, a sample recombination strategy is designed to increase the variety of classification tasks in the training period. Second, an embedding module is employed to extract deep features of the input image patches. Third, we propose two relation modules: image-to-class (I2C) block and image-to-image (I2I) block. In the I2C block, the second-order features between support samples and query samples are generated, and then, the I2C relation score is calculated for the second-order features. In the I2I block, the first-order features of support samples and query samples are connected by concatenation, difference, and product, and then, the I2I relation score is calculated for the first-order features. Finally, DM-MRN is constructed by one embedding module, two I2C blocks, and one I2I block. Three different features are first yielded by the embedding module, which are then fed into the relation module to calculate different relation scores. Moreover, an adaptive weighting strategy is designed to fuse the relation scores, and the class for each query sample can be assigned according to the highest value of the fused relation score.

The main innovative contributions of this work can be summarized as follows.

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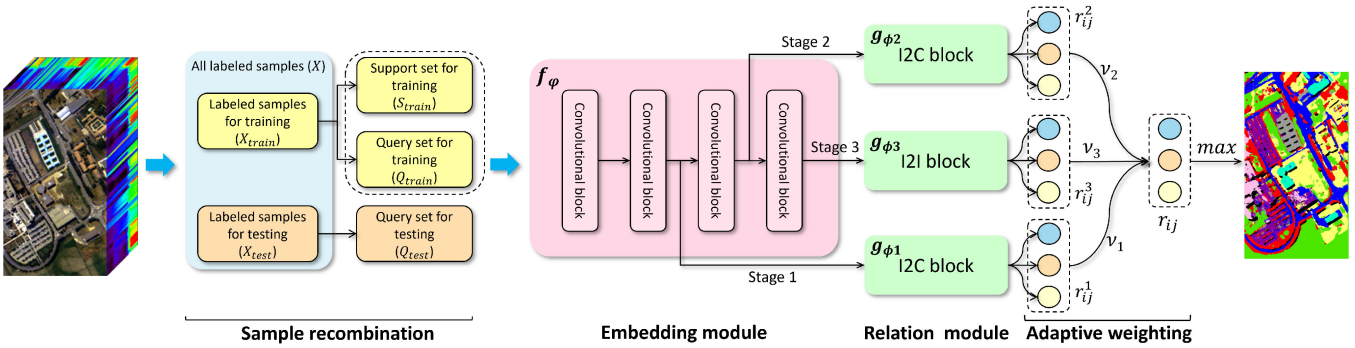


Fig. 1. Flowchart of the proposed DM-MRN for few-shot HSI classification.

TABLE I
DETAILS OF THE DM-MRN ARCHITECTURE

		Convolution	Filter size	BN	ReLU	Max-pooling
Embedding module		1	$n_c \times 64 \times 1 \times 1$	64	Y	2×2
		2	$64 \times 64 \times 1 \times 1$	64	Y	N
		3	$64 \times 64 \times 1 \times 1$	64	Y	N
		4	$64 \times 64 \times 1 \times 1$	64	Y	N
Relation module	Image-to-class (I2C) metric module	5	$1 \times 64 \times 1 \times n_r$	64	Y	N
		6	$64 \times 64 \times 3 \times 3$	64	Y	N
		7	$64 \times 1 \times 3 \times 3$	N	N	N
	Image-to-image (I2I) metric module	8	$256 \times 128 \times 1 \times 1$	128	Y	N
		9	$128 \times 64 \times 1 \times 1$	64	Y	N
		10	$64 \times 64 \times 1 \times 1$	64	Y	N
		11	$64 \times 1 \times w \times w$	N	N	N

1) DM-MRN is proposed for few-shot HSI classification which is constructed by a multistage structure with two metrics at the same time.
2) A novel I2C block is designed, which can generate second-order features between query samples and support samples by using deep local descriptors. Nevertheless, the I2C block can condense class-level features with limited training samples.
3) A novel I2I block is conceived, which can simultaneously learn feature relations from the channel and spatial dimensions by combining feature concatenation, feature difference, and feature product.

II. PROPOSED METHODOLOGY

An overall structure of the proposed framework is illustrated in Fig. 1. First, sample recombination is designed to generate different input instances before each training iteration. Second, the embedding module is used to generate features for three stages. Third, the relation module is constructed by two novel metric blocks (I2C and I2I blocks). Finally, adaptive weighting is employed to fuse different metric scores for classification. Table I shows more details of DM-MRN architecture.

A. Sample Recombination

To increase the variety of classification tasks in the training period, we design a sample recombination strategy, which is used to generate different input instances containing support sets and query sets before each training iteration.

Considering that there are a total of N labeled samples (including C classes) in the HSI dataset, we denoted all labeled samples as $X = \{X_i\}_{i=1}^{N}$. First, s labeled samples from each class ($s \times C$ samples in total) in X are randomly selected to form the training set X_{train}, and all samples in the rest of X form the testing set X_{test}. According to (1), if we want to maximize the combination number, we need to make n close to $(m/2)$. Therefore, before each training iteration, half of the samples (written as b) in X_{train} are randomly selected as the training support set $S_{\text{train}} = \{s_{ij}\}$, and one sample of each class in X_{train} is randomly selected as the training query set $Q_{\text{train}} = \{q_j\}$. S_{train} and Q_{train} together compose an input instance M for the model, as shown in (2)

$$C_m^n = \frac{m!}{n!(m-n)!} \cdot (n \le m) \tag{1}$$

$$M = [s_{11}, \ldots, s_{b1}, \ldots, s_{1C}, \ldots, s_{bC}, q_1, \ldots, q_c]. \tag{2}$$

With the similar construction of M, we construct an input instance M' for the testing phase. All samples in the testing set X_{test} becomes testing query sample (i.e., $Q_{\text{test}} \equiv X_{\text{test}}$), and all samples in the training set X_{train} become testing support samples (i.e., $S_{\text{test}} \equiv X_{\text{train}}$).

B. Embedding Module

After getting an input instance M, an embedding module consisting of four convolutional blocks is employed to extract the deep features of the patch data. Each block is the concatenation of a 2-D convolution filter, a batch normalization layer, and a ReLU activation layer. More details of the embedding module are listed in Table I.

The structure of each convolutional block is almost the same, except for the first convolutional block which contains

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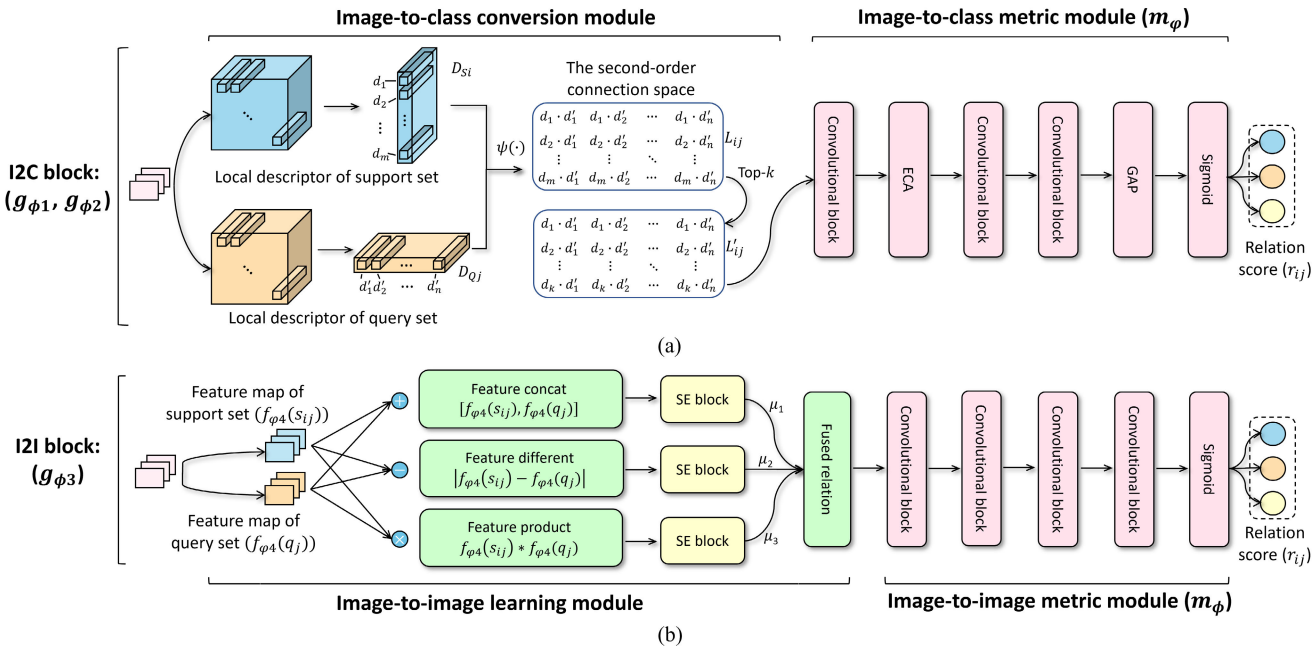


Fig. 2. Architecture of relation module. (a) I2C block. (b) I2I block.

an additional 2×2 max-pooling layer and the filter number is $n_C \times 64$. The first convolutional block aims to map the input instance to 64-D deep features. Whereas, the function of the latter three convolutional blocks is to further extract the different features of input instances. Since the embedding module merely performs some linear combination operations, the generated features can be regarded as first-order features.

C. Relation Module

1) Image-to-Class Block: The original relation network uses image-level global features which are difficult to represent the class distribution effectively under limited samples. Therefore, we designed an I2C block that can condense class-level features through limited samples to fill the single metric. Fig. 2(a) shows the composition of the I2C block, which can be divided into the I2C conversion module and I2C metric module.

1) *Image-to-Class Conversion Module:* This module aims at expressing deep features that yield from the embedding module in the form of local descriptors and constructing the second-order connection space. Then, we further simplify this space through Top-k.[1] Specifically, the deep features $f_{\varphi(t+1)}(s_{ij})$ and $f_{\varphi(t+1)}(q_j)$ are generated by embedding module. As mentioned in [51], the deep features of $h \times w \times d$ can be expressed as K ($K = h \times w$) d-dimensional local descriptors. Therefore, we can regard $f_{\varphi(t+1)}(s_{ij})$ as m ($m = b \times h \times w$) local descriptors D_{Si} of each class, and $f_{\varphi(t+1)}(q_j)$ as n ($n = 1 \times h \times w$) local descriptors D_{Qj} of each class. Subsequently, D_{Si} and D_{Qj} generate a second-order connection space L_{ij} through ψ, which can be expressed by the following:

$$\begin{aligned} D_{\mathrm{Si}} &= [d_{i1}, \ldots, d_{\mathrm{im}}], \quad i = \{1, \ldots, C\} \\ D_{Qj} &= [d'_{j1}, \ldots, d'_{\mathrm{jn}}], \quad j = \{1, \ldots, C\} \qquad (3) \\ L_{ij} &= \psi({D_{\mathrm{Si}}}^T, D_{Qj}) \\ &= {D_{\mathrm{Si}}}^T \otimes D_{Qj}, \quad i, j = \{1, \ldots, C\}. \qquad (4) \end{aligned}$$

The second-order connection space L_{ij} is the product of D_{Si}^T and D_{Qj}. Therefore, we can regard the connection space as a feature formed by linear–nonlinear combination operation, that is, a second-order feature. Each element in L_{ij} is the dot product between the query sample and support samples of a certain class. According to (5), the dot product reflects the similarity of two vectors. Consequently, we can determine whether the query sample belongs to a certain class in the support set by learning the similarity of L_{ij} as

$$\vec{a} \cdot \vec{b} = \sum_{i=1}^{n} a_i b_i = |\vec{a}||\vec{b}| \cos\theta. \qquad (5)$$

However, we also consider that the connection space is redundant. Therefore, before performing the I2C metric, k largest elements are selected along the column in L_{ij} to form the simplified connection space L'_{ij}.

2) *Image-to-Class Metric Module (m_φ):* Instead of using the traditional distance metric function, this module is designed to learn the similarity of the simplified connection space L'_{ij}. More details of the I2C metric module are listed in Table I.

First, the convolutional block with the adaptive convolutional kernel ($1 \times n_r$) is used to reduce the size of L'_{ij} from $k \times n$ to $k \times k$. Second, an efficient

[1]Given an input tensor, Top-k returns the k largest elements along a given dimension.

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channel attention (ECA) [52] block is used to focus the key information in the feature maps. Third, the convolutional block with 3×3 convolutional kernel further extracts the features in L'_{ij}. Finally, a relation score r^t_{ij} $(t = 1, 2)$ related to L'_{ij} is generated by a 2-D convolution with 3×3 convolutional kernel, a global average pooling (GAP) [53] layer and a sigmoid activation layer. Equation (6) formulated the generation of r^t_{ij}

$$\begin{aligned} r^t_{ij} &= g_{\phi t}(f_{\varphi(t+1)}(s_{ij}), f_{\varphi(t+1)}(q_j)) \\ &= m_\varphi[\text{Top-}k(L_{ij})] \end{aligned} \tag{6}$$

where $g_{\phi t}$ $(t = 1, 2)$ represents I2C block, t is the corresponding stage of $g_{\phi t}$, and m_φ is the I2C metric module in I2C block.

In our proposed DM-MRN, there are two identical I2C blocks ($g_{\phi 1}$ and $g_{\phi 2}$) with unshared parameters. The reason is that we need two I2C blocks to separately process the features generated from the second and third convolutional blocks in the embedding module.

2) Image-to-Image Block: The relation network [47] merely relies on feature concatenation for similarity learning. Besides, it mainly focuses on the feature relation in the channel dimension, while ignoring the spatial dimension. Therefore, inspired by the multirelation attention learning network (MRAN) [54], we improve the I2I metric and propose the I2I block. Fig. 2(b) shows the composition of the I2I block, which can be divided into the I2I learning module and the I2I metric module.

1) *Image-to-Image Learning Module:* To leverage the channel and spatial information, we fuse various feature relations between $f_{\varphi 4}(s_{ij})$ and $f_{\varphi 4}(q_j)$ by combining feature concatenation (f_c), feature difference (f_d), and feature product (f_p). The three feature relations take the form

$$\begin{aligned} f_c &= f_{\varphi 4}(s_{ij}) \oplus f_{\varphi 4}(q_j) \\ f_d &= |f_{\varphi 4}(s_{ij}) - f_{\varphi 4}(q_j)| \\ f_p &= f_{\varphi 4}(s_{ij}) \otimes f_{\varphi 4}(q_j) \end{aligned} \tag{7}$$

where $A \oplus B$ means that A and B are concatenating along channel, $|A - B|$ denotes the absolute difference between A and B (elementwise subtraction), and $A \otimes B$ represents product between A and B (elementwise multiplication). We first strengthen feature relations by a squeeze-and-excitation (SE) block [55], and then, the strengthened feature relations are adaptively fused. Particularly, f_c, f_d, and f_p are fed into the SE block, respectively, and new features f'_c, f'_d, and f'_p are generated. The fused feature f_{Fused} can be formulated as follows:

$$f_{\text{Fused}} = \left[\mu_1 \times f'_c, \mu_2 \times f'_d, \mu_3 \times f'_p\right] \tag{8}$$

$$\mu'_i = \mu_i \Big/ \sum_{j=1}^{3} \mu_j, \quad (i = 1, 2, 3) \tag{9}$$

where μ_1, μ_2, and μ_3 are the adaptive weights, whose initial values are set to 1/3. Moreover, the normalized weights μ'_i is generated by (9) before each fusion.

2) *Image-to-Image Metric Module (m_ϕ):* This module mainly consists of four convolutional blocks. More details of the I2I metric module are listed in Table I. First, three convolutional blocks with 1×1 convolutional kernel further extract the features of f_{Fused} in different channel dimensions. Then, a relation score r^3_{ij} related to f_{Fused} is generated by a 2-D convolution with $w \times w$ convolutional kernel and a sigmoid activation layer, where w is patch size of the input tensor. Equation (10) formulated the generation of r^3_{ij}

$$\begin{aligned} r^3_{ij} &= g_{\phi 3}(f_{\varphi 4}(s_{ij}), f_{\varphi 4}(q_j)) \\ &= m_\phi(f_{\text{Fused}}) \end{aligned} \tag{10}$$

where $g_{\phi 3}$ represents I2I block, and m_ϕ is the I2I metric module in I2I block.

D. Adaptive Weighting and Classification

As we can see from Fig. 1, the relation module of DM-MRN contains three stages, which means that three relation scores are generated from the relation module. Considering that different metrics have different performances in feature separability, we design an adaptive weighting strategy to fuse the relation scores r^1_{ij}, r^2_{ij}, and r^3_{ij}. This process can be summarized as (11)

$$r^f_{ij} = \nu_1 \times r^1_{ij} + \nu_2 \times r^2_{ij} + \nu_3 \times r^3_{ij} \tag{11}$$

$$\nu'_i = \nu_i \Big/ \sum_{j=1}^{3} \nu_j, \quad (i = 1, 2, 3) \tag{12}$$

where ν_1, ν_2, and ν_3 are the adaptive normalized weights corresponding to relation scores r^1_{ij}, r^2_{ij}, and r^3_{ij}, and the normalization formula is listed in (12). In addition, ν_1, ν_2, and ν_3 all have an initial value of one third.

Furthermore, we classify the query samples by r_{ij}. r_{ij} is a 1-D array, and each value in this array represents the similarity between the query sample and a certain class. By comparing the values in the array, we can assign a class label corresponding to the maximum value to the sample and then complete the classification. The class label of the query sample can be obtained by the following:

$$\text{Label}(q_j) = \arg\max_j \Big\{ \max_i \left\{ r^f_{ij} \right\} \Big\}. \tag{13}$$

E. Loss Function

Although the proposed model is used for classification, the predicted result is a relation score in the range of [0,1]. Therefore, mean squared error (MSE) loss is employed as the loss function for our model, which can be defined by the following:

$$\ell = \frac{1}{C} \sum_{j=1}^{C} \left(\text{Label}(q_j) - y_j\right)^2 \tag{14}$$

where C represents the number of classes.

As mentioned before, we have obtained four relation scores r^1_{ij}, r^2_{ij}, r^3_{ij}, and r^f_{ij}. In addition, three single feature relation in I2I block can also generate three relation scores r^c_{ij}, r^d_{ij}, and

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TABLE II
DETAILS OF THE FIVE DATASETS

	University of Pavia	Salinas	Yellow River Delta	Berlin	Houston2018
Region or country	Pavia	California	China	Berlin	Houston
Sensor type	ROSIS	AVIRIS	AHSI	HyMap	ITRES CASI 1500
Number of bands	103	204	119	244	48
Wavelength (μm)	0.43-0.86	0.40-2.50	0.40-2.50	0.40-2.50	0.38-1.05
Spatial resolution (m)	1.3	3.7	30	30	1.0
Image size	610×340	512×217	1147×1600	1723×476	601×2384
Number of labeled samples	42776	54129	9825	464671	504712
Number of classes	9	16	23	8	20

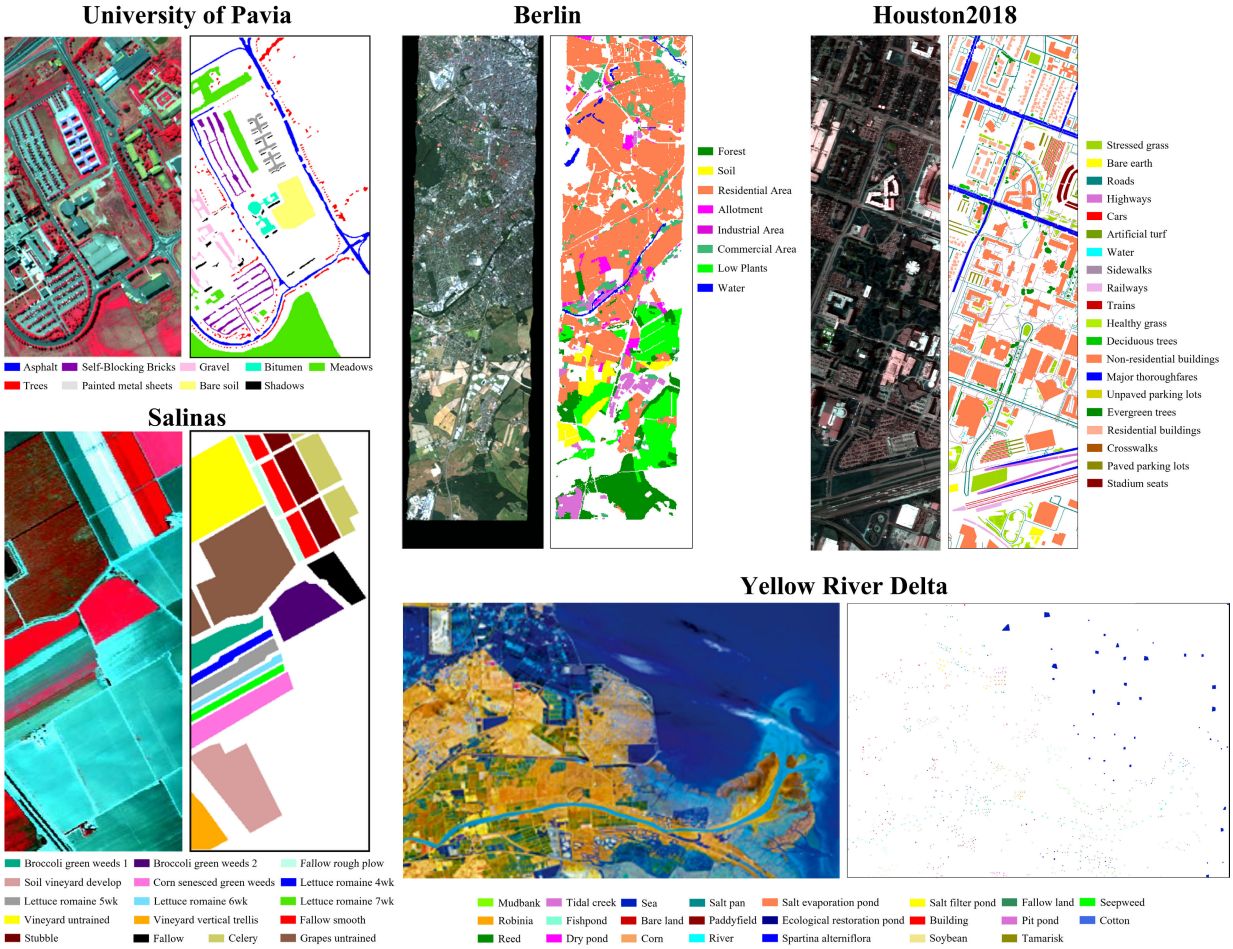


Fig. 3. False-color composite images and the ground-truth maps on the five considered datasets, i.e., University of Pavia, Salinas, Yellow River Delta, Berlin, and Houston2018.

r_{ij}^{p}. Therefore, we can get seven corresponding losses, which can be divided into three parts: the first part is feature relation loss ℓ_{fr}, the second part is stage loss ℓ_{stage}, and the last part is fused loss ℓ_{fused}. The total loss of the proposed method can be formulated as

$$L = \ell_{fr} + \ell_{\text{stage}} + \ell_{\text{fused}} \tag{15}$$

where ℓ_{fr} denotes the sum of losses corresponding to r_{ij}^{c}, r_{ij}^{d}, and r_{ij}^{p}, ℓ_{stage} represents the sum of losses corresponding to r_{ij}^{1}, r_{ij}^{2}, and r_{ij}^{3}, and ℓ_{fused} is the loss corresponding to r_{ij}^{f}. We consider that ℓ_{fr} plays a greater role in L, which is supported by the experimental results in Section III-E.

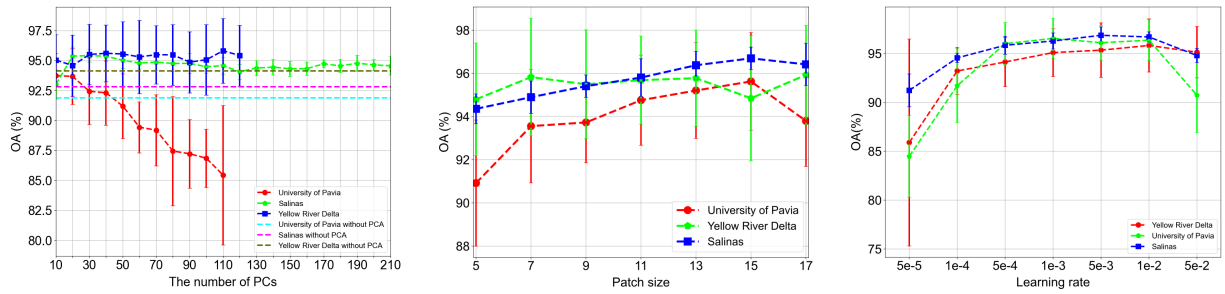
Moreover, we use Adam optimizer [56] to calculate and update the network parameters, while minimizing the loss L.

III. EXPERIMENTAL RESULTS

A. Hyperspectral Datasets

We select the popular datasets including the University of Pavia and Salinas and novel datasets including Yellow River Delta, Berlin, and Houston2018 to verify the proposed method. Among them, the novel datasets Berlin and Houston2018 are mainly used to evaluate the HSI classification performance and ablation study. Specific details of each dataset are shown in Table II. In addition, Fig. 3 shows the false color images and the corresponding ground-truth of five datasets.

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(a) (b) (c)

Fig. 4. Evolution of OA as a function of (a) number of PCs, (b) patch size, and (c) learning rate.

B. Experiment Settings

1) For Hyperparameters: The training iteration is set to 1000 for all datasets. In addition, other parameters, including learning rate, reduced-dimension by PCA, patch size, and k value in Top-k, are analyzed in the experiment of hyperparameter analysis.

2) For Sample Selection: We randomly select ten labeled samples per class for training, while the rest of the labeled samples are used for testing. However, for the hyperparameter analysis experiment, we randomly select 20 labeled samples per class, and for the generalization performance experiment, we set the labeled samples per class in the range of [1, 2, 3, 4, 5, 10, 15, 20, 25, 30]. If there are so few labeled samples for a certain class that the selected samples are less than half of the total labeled samples, we randomly divide the labeled samples of this class into two halves, one half for training and the other half for testing.

3) For Performance Comparison: We compare the proposed method with three common DL methods, five popular FSL methods, and two state-of-the-art (SOTA) models, which are 2-D-CNN [20], 3-D-CNN [20], spectral–spatial residual network (SSRN) [21], DPN-HRA [43], similarity-based deep metric model (S-DMM) [57], FSL framework with a class-covariance metric (CMFSL) [58], spectral–spatial siamese network (S3Net) [41], deep cross-domain few-shot learning (DCFSL) [59], SSFTT [27], and minibatch GCN (miniGCN) [33]. For DPN-HRA, S-DMM, CMFSL, S3Net, DCFSL, SSFTT, and miniGCN, the hyperparameters are set as the recommended values. As for 2-D-CNN, 3-D-CNN, and SSRN, the patch size is set to 9×9, 5×5, and 7×7, respectively, and other parameters are also set as the recommended values.

4) Running Platforms and Evaluation Metrics: All the experiments are implemented by Pytorch 1.9.0 on a Windows 11 platform with 12th generation Core i5 CPU, RTX 3060 GPU, and 16-GB RAM. The evaluation metrics used in the experiments include overall accuracy (OA), average accuracy (AA), and Kappa coefficient (κ). To increase the stability and convincing of the experimental results, we conduct ten independent runs for each experiment and report the averaged results. The source code will be available online (https://github.com/ZhaohuiXue/DM-MRN).

TABLE III
IMPACT OF k (TOP-k) ON OA

Methods	University of Pavia	Salinas	Yellow River Delta
$k = 1$	95.17±3.12	95.51±0.88	94.42±2.38
$k = 5$	**96.55±2.05**	**96.88±0.85**	**95.83±2.72**
$k = 9$	96.15±2.27	96.03±0.96	94.91±2.61
$k = 20$	96.06±2.33	95.69±0.79	94.70±3.24

C. Hyperparameter Analysis

1) Reduced-Dimension: To select the optimal reduced-dimension of PCA, Fig. 4(a) shows the evolution of OA with the number of PCs. According to the curve for the University of Pavia dataset, we can easily find that the OA using PCA (when the number of PCs is lower than 40) is higher than that without PCA. However, for Salinas and Yellow River Delta datasets, using PCA can always get higher OAs. According to the highest OA, we set the optimal reduced-dimension to 10, 30, and 30 for the three datasets.

2) Patch Size: To choose the optimal patch size, we set it in the range of [5×5, 7×7, 9×9, 11×11, 13×13, 15×15, 17×17]. Fig. 4(b) shows the evolution of OA with the patch size. As shown in the figure, we can easily find that the proposed model can achieve the highest OA in the University of Pavia and Salinas datasets when the patch size is set to 15×15. However, for the Yellow River Delta dataset, OA is generally insensitive to patch size, but the highest OA occurs when the patch size is set to 7×7. In this context, we set the patch size to 15×15, 15×15, and 7×7 for the three datasets.

3) Learning Rate: To find the optimal initial learning rate, we set it in the range of [0.05, 0.01, 0.005, 0.001, 0.0005, 0.0001, 0.00005]. Fig. 4(c) shows the evolution of OA with the learning rate. As we can see, OA is significantly impacted by the learning rate for the three datasets, and the optimal learning rate of the three datasets are 0.001, 0.005, and 0.01.

4) Top-k: In order to examine the influence of different k on space L_{ij} for fitting similarity, we set it in the range of [1], [5], [9], and [20]. As shown in Table III, the averaged OA is relatively low when k is too small or too large (i.e., $k = 1$ or $k = 20$), and the highest averaged OA is obtained when the value of k is set to 5.

D. Generalization Performance

To examine the generalization performance of the proposed method, we select a few labeled samples from each class, i.e.,

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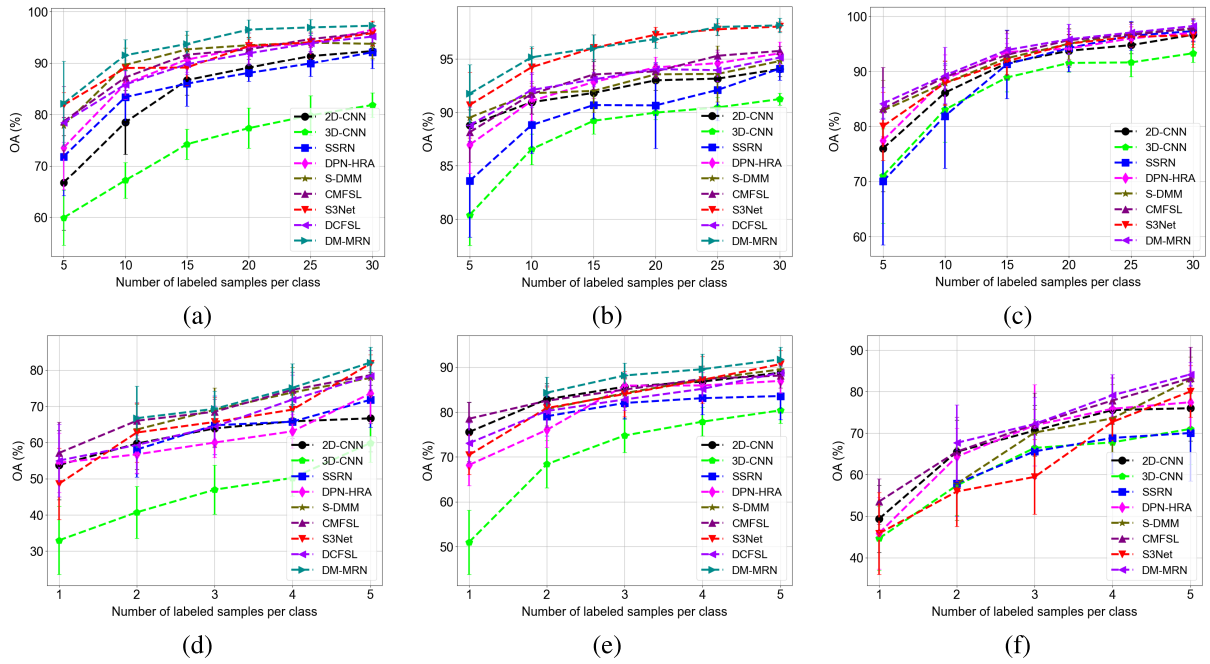


Fig. 5. Evolution of OA as a function of the number of labeled samples per class. (a) 5–30 for the University of Pavia, (b) 5–30 for Salinas, (c) 5–30 for Yellow River Delta, (d) 1–5 for the University of Pavia, (e) 1–5 for Salinas, and (f) 1–5 for Yellow River Delta.

[5, 10, 15, 20, 25, 30]. Fig. 5(a)–(c) shows the evolution of OA with a different number of labeled samples per class. DM-MRN achieves the highest OA under all conditions on three datasets, which demonstrates the superiority of DM-MRN in generalization performance.

Moreover, the classification performance of all methods under extreme conditions (1–4 training samples per class) is also examined. Fig. 5(d) and (e) summarizes the classification results. In order to improve the stability of the model under extremely few training samples, we reported the average results of 20 independent runs of each experiment. It is worth noting that SSRN, S-DMM, and DM-MRN cannot run under one training sample per class due to the model structure. As we can see, FSL methods perform better in scarce training samples than common DL methods, and DM-MRN stands out from the compared FSL methods. Specifically, DM-MRN can always achieve higher OA under all conditions on three datasets, which proves that the proposed method still has advantages under extreme conditions.

E. Ablation Study

1) Effectiveness of Sample Recombination: To examine the performance improvement of sample recombination, we compare two variants of DM-MRN: one is DM-MRN (without sample recombination), which does not divide X_{train} into support set and query set by the strategy of maximizing the number of combinations, the other one is DM-MRN (Ours). As we can see in Fig. 6, the averaged OA of DM-MRN (Ours) is higher than that of DM-MRN (without sample recombination). Specifically, the classification accuracy of the model without sample recombination decreased by 3.06%–13.61% in three datasets. This is due to the model can obtain more generalized classification knowledge through the variety of input instances generated by the sample recombination.

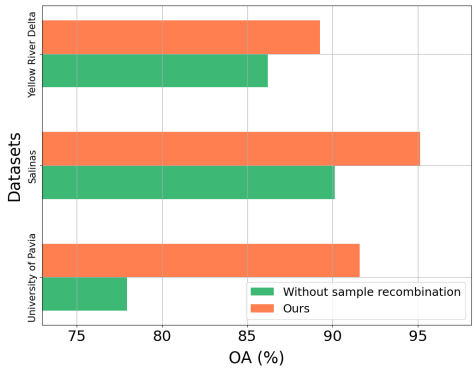


Fig. 6. Ablation analysis of sample recombination of DM-MRN.

2) Effectiveness of Multistage Structure: To verify the effectiveness of multistage structure, we compare four variants of DM-MRN: DM-MRN (only with Stage 1), DM-MRN (only with Stage 2), DM-MRN (only with Stage 3), and DM-MRN (Ours). The first three variants only remain one block in the relation module, but the last one is a multistage network whose relation module consists of three blocks. As we can see in Table IV, the averaged OA obtained by DM-MRN (Ours) is always higher than that obtained by the other three, which proves the advantage of the multistage structure. Specifically, the classification accuracy of the model without a multistage structure decreased by 0.40%–4.50% in three datasets.

3) Effectiveness of Dual-Metric: To validate the effectiveness of dual-metric, we compare three variants of DM-MRN: DM-MRN (only with I2C block), DM-MRN (only with I2I block), and DM-MRN (Ours). The first two variants only remain one kind of metric block, and the third variant uses a dual-metric block. As we can see in Table V, the averaged

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TABLE IV
ABLATION ANALYSIS OF MULTISTAGE STRUCTURE OF DM-MRN

Stage 1	Stage 2	Stage 3	University of Pavia	Salinas	Yellow River Delta
✓	×	×	88.24	93.33	86.69
×	✓	×	89.40	92.85	84.76
×	×	✓	90.45	94.72	88.80
✓	✓	✓	**91.57**	**95.12**	**89.26**

TABLE V
ABLATION ANALYSIS OF DUAL-METRIC OF DM-MRN

I2C	I2I	University of Pavia	Salinas	Yellow River Delta
✓	×	89.52	93.48	87.82
×	✓	90.45	94.72	88.80
✓	✓	**91.57**	**95.12**	**89.26**

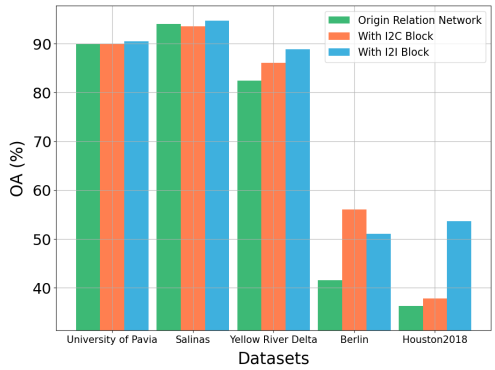


Fig. 7. Ablation analysis of I2C block and I2I block of DM-MRN.

OA of DM-MRN (Ours) is always higher than that of others in the three datasets. Specifically, the classification accuracy of the model without dual-metric decreased by 0.40%–2.05% in three datasets. Therefore, we conclude that dual-metric indeed improves the classification performance.

4) Effectiveness of I2C Block and I2I Block: To examine the effectiveness of the I2C block and I2I block, we compare three variants of DM-MRN: DM-MRN (only with the original relation module), DM-MRN (replace the original relation module with I2C block), and DM-MRN (replace the original relation module with I2I block). As we can see in Fig. 7, the model with I2C block or I2I block can always achieve higher OA than the original relation network, which proves the effectiveness of the proposed I2C block and I2I block. Specifically, the classification accuracy of the model without the I2C block decreased by 0.03%–14.50%, and without the I2I block decreased by 0.49%–17.35%.

5) Importance of Loss Function: To compare which part of the total loss L plays a greater role, we designed three experiments, which are DM-MRN without ℓ_{fr}, ℓ_{stage}, and ℓ_{fused}. As we can see in Table VI, compared with our proposed model, the lack of any part of L leads to the degradation of classification performance. Specifically, the classification accuracy of DM-MRN without ℓ_{fr}, ℓ_{stage}, and ℓ_{fused} on three datasets decreased by 1.56%, 0.85%, and 0.62% at the highest. It can be seen that ℓ_{fr} has a greater influence on the training of DM-MRN. We can explain this phenomenon from two perspectives. From the model structure, ℓ_{fr} is located in the middle of the DM-MRN, which plays a key role in the training

TABLE VI
ABLATION ANALYSIS OF LOSS FUNCTIONS OF DM-MRN

Methods	University of Pavia	Salinas	Yellow River Delta
Without ℓ_{fr}	89.98±3.60	94.57±1.19	88.61±2.03
Without ℓ_{stage}	90.72±3.80	94.82±0.62	89.05±3.43
Without ℓ_{fused}	90.95±3.42	94.91±0.83	89.19±2.18
Ours	**91.57±3.49**	**95.12±0.85**	**89.26±5.10**

and prediction of the model. From the backpropagation, the adjustment of the weight of each layer will be affected by the subsequent layers, which leads to the problem that the weight update of the front part of the model is weakened. Therefore, the importance of ℓ_{fr} to the DM-MRN training phase is highlighted.

F. Classification Results

In this section, we randomly select ten labeled samples from each class for training to compare the performance of 2-D-CNN, 3-D-CNN, SSRN, DPN-HRA, S-DMM, DCFSL, CMFSL, S3Net, SSFTT, miniGCN, and DM-MRN. It is noted that DCFSL can only distinguish 19 classes at most, so this model cannot be evaluated by the Yellow River Delta dataset (23 classes) and the Houston2018 dataset (20 classes).

1) University of Pavia: According to the results reported in Table VII, the accuracies of DM-MRN and S-DMM are obviously higher than the other methods. We consider that this is because these two networks are both designed based on relation network, which has stronger feature learning ability than others with few training samples. Specifically, DM-MRN achieves the highest averaged OA of 91.57%, which increases from 1.94% to 27.98% compared with other methods. Besides, the AA and κ of DM-MRN are also obviously improved. As shown in Fig. 8, the classification map of DM-MRN is obviously less noisy and smoother. It is worth noting that the classification map of DM-MRN is the smoothest in *Meadows* and *Bare soil*, while other methods have some misclassified pixels. At the same time, DM-MRN can accurately classify test samples in unlabeled areas.

2) Salinas: According to the results given in Table VIII, the accuracies of DM-MRN and S3Net are obviously higher than other methods. Specifically, DM-MRN achieves the highest averaged OA of 95.12%, which increases from 0.88% to 13.14% compared with other methods. Similar observations can be found for AA and κ. In terms of class-specific accuracy, DM-MRN has the highest averaged OA among the seven classes, which demonstrates excellent classification performance. As shown in Fig. 9, DM-MRN produces a smoother classification map, which helps to achieve intraclass aggregation and interclass separation. It is worth noting that DM-MRN achieves the highest accuracy in classifying *Vineyard untrained*, whereas other methods mistakenly identify some pixels.

3) Yellow River Delta: According to the results listed in Table IX, the accuracies of DM-MRN, DPN-HRA, and CMFSL are obviously higher than other methods. Specifically, DM-MRN achieves the highest averaged OA with 89.26%, which increases from 0.30% to 7.37% compared to other methods. In terms of class-specific accuracy, DM-MRN has

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TABLE VII

CLASSIFICATION ACCURACIES OBTAINED BY DIFFERENT METHODS FOR UNIVERSITY OF PAVIA (USING TEN LABELED SAMPLES PER CLASS FOR TRAINING)

Class	2D-CNN [20]	3D-CNN [20]	SSRN [21]	DPN-HRA [43]	S-DMM [57]	CMFSL [58]	S3Net [41]	DCFSL [59]	SSFTT [27]	miniGCN [33]	DM-MRN (Ours)
1	93.49±3.86	74.90±8.37	92.04±9.26	94.54±4.11	**94.85±3.31**	81.09±10.07	89.02±6.90	85.68±6.17	83.30±5.70	48.08±9.87	92.08±5.65
2	96.64±1.76	87.73±2.49	95.92±3.07	**97.53±1.75**	84.55±9.33	88.08±7.26	84.84±4.79	87.18±4.60	81.20±7.74	59.13±13.73	89.31±8.47
3	56.84±12.24	42.01±7.41	63.84±19.45	67.54±10.29	**89.56±3.69**	81.51±2.82	84.55±6.14	70.70±13.68	79.30±8.43	71.14±11.05	83.99±8.26
4	75.86±16.41	67.25±7.14	77.95±18.91	84.42±8.49	**96.07±3.31**	94.25±3.24	92.75±5.16	95.70±1.05	87.76±4.20	89.11±6.67	95.20±3.34
5	98.81±1.49	99.61±0.67	99.43±0.79	97.81±2.00	**100.00±0.0**	99.98±0.07	99.81±0.34	99.90±0.19	99.51±0.43	99.16±0.40	99.99±0.02
6	64.47±12.40	50.46±10.93	70.01±9.30	66.98±10.67	88.71±11.28	88.91±10.63	95.08±5.07	84.23±8.17	90.81±7.54	69.86±10.12	**97.08±2.27**
7	58.37±19.46	47.13±5.47	72.15±20.86	82.00±14.64	99.66±0.66	88.30±5.98	99.73±0.32	86.41±7.08	98.09±1.19	90.02±4.40	**99.76±0.21**
8	74.76±7.80	49.52±9.44	77.00±11.04	78.51±7.65	**91.86±3.88**	80.40±8.25	91.41±5.40	73.01±11.07	83.66±6.19	48.34±17.42	88.06±6.09
9	57.96±13.59	58.82±15.33	99.37±0.86	92.88±4.72	**99.98±0.04**	99.81±0.24	97.36±1.93	99.64±0.49	95.47±4.32	99.96±0.05	98.72±1.01
OA (%)	78.48±6.18	67.16±3.47	83.37±5.18	85.95±3.04	89.63±3.34	87.19±3.42	89.02±2.52	85.83±2.70	84.65±3.56	63.59±6.27	**91.57±3.49**
AA (%)	75.25±3.51	64.16±2.44	83.08±4.56	84.69±2.78	**93.92±1.03**	89.15±1.89	92.73±1.33	86.94±1.94	88.79±1.61	74.98±2.81	93.80±1.39
$\kappa\times100$	72.95±7.16	58.67±3.95	78.72±6.26	81.94±3.73	86.66±4.03	83.43±4.23	85.87±3.12	81.58±3.40	80.38±4.26	55.24±6.31	**89.11±4.28**

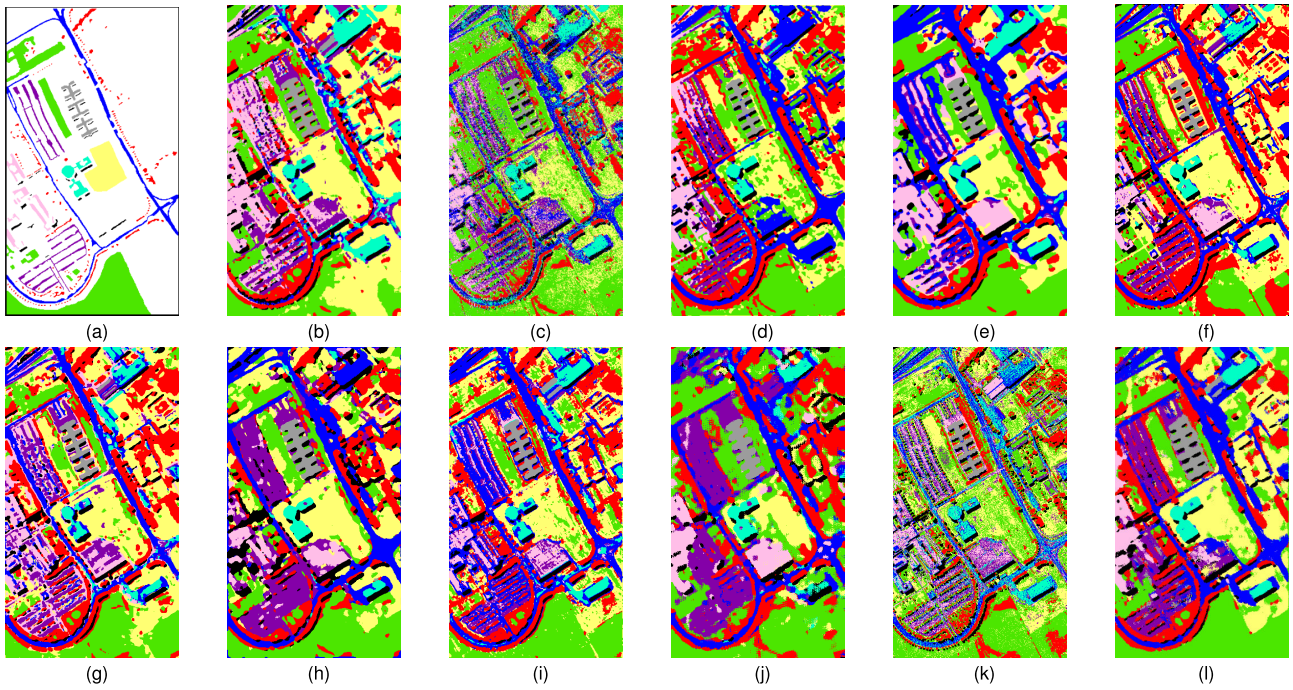


Fig. 8. Classification maps obtained by different methods for the University of Pavia. (a) Ground-truth, (b) 2-D-CNN (78.48%), (c) 3-D-CNN (67.16%), (d) SSRN (83.37%), (e) DPN-HRA (85.95%), (f) S-DMM (89.63%), (g) CMFSL (87.19%), (h) S3Net (89.02%), (i) DCFSL (85.83%), (j) SSFTT (84.65%), (k) miniGCN (63.59%), and (l) DM-MRN (91.57%).

TABLE VIII

CLASSIFICATION ACCURACIES OBTAINED BY DIFFERENT METHODS FOR SALINAS (USING TEN LABELED SAMPLES PER CLASS FOR TRAINING)

Class	2D-CNN [20]	3D-CNN [20]	SSRN [21]	DPN-HRA [43]	S-DMM [57]	CMFSL [58]	S3Net [41]	DCFSL [59]	SSFTT [27]	miniGCN [33]	DM-MRN (Ours)
1	99.31±1.12	97.53±4.94	98.45±4.64	99.24±2.20	99.04±1.02	99.39±0.76	99.61±0.46	99.59±0.96	99.44±0.93	99.18±0.31	**99.91±0.12**
2	99.03±1.03	98.06±1.33	**99.99±0.02**	99.88±0.14	99.78±0.17	99.88±0.26	96.03±7.80	99.85±0.19	98.75±0.76	95.05±3.16	99.78±0.35
3	95.91±3.09	89.91±5.13	96.37±4.61	98.37±1.64	97.38±4.57	99.85±0.16	99.94±0.13	99.40±0.99	98.30±1.73	76.87±7.90	**99.98±0.04**
4	94.50±3.04	94.98±3.33	97.61±1.77	93.49±3.66	99.10±1.33	**99.52±0.55**	98.97±1.16	99.35±0.97	94.12±6.18	99.09±0.40	97.48±5.21
5	91.02±8.70	94.21±3.39	95.45±5.01	98.47±0.99	**98.55±2.41**	94.28±2.59	97.86±3.58	95.51±2.47	87.11±11.45	95.29±3.03	97.11±2.04
6	98.69±1.67	99.84±0.24	99.93±0.10	98.69±1.76	**99.97±0.06**	98.21±1.76	99.66±0.41	99.52±0.64	95.71±5.12	99.70±0.13	99.25±0.97
7	98.17±2.77	97.22±2.93	98.91±1.13	99.02±1.13	**99.97±0.03**	99.30±0.57	99.90±0.26	99.39±0.55	95.17±4.12	99.29±0.37	99.08±1.59
8	87.20±3.03	78.53±3.07	83.05±7.48	88.61±6.55	75.68±8.71	79.36±8.73	87.55±4.53	80.94±5.41	74.89±6.39	49.37±17.70	**88.67±2.76**
9	99.03±0.87	96.63±1.81	99.43±0.63	99.23±0.50	99.29±1.26	98.47±2.00	99.83±0.30	99.73±0.38	98.31±1.60	99.30±0.47	**99.97±0.05**
10	89.90±2.60	90.81±4.67	92.39±3.13	94.18±3.13	95.02±3.27	90.72±4.56	96.00±4.16	92.64±3.22	92.77±1.38	86.71±5.10	**96.00±4.58**
11	79.34±7.32	93.65±2.93	95.70±1.33	86.41±10.30	98.37±1.99	98.39±2.50	99.40±0.89	98.82±1.61	96.90±3.26	93.33±2.49	**99.87±0.19**
12	98.86±0.89	89.12±5.71	98.98±0.94	96.86±4.65	**99.90±0.30**	99.82±0.28	94.78±8.47	99.89±0.12	86.59±7.72	97.83±4.43	98.47±1.34
13	99.53±0.46	80.71±7.09	98.86±0.84	95.60±6.38	99.36±0.75	99.02±0.96	98.09±2.51	**99.60±0.28**	90.04±10.91	96.72±4.21	98.02±2.87
14	91.85±7.47	79.66±8.59	96.62±5.34	91.57±11.64	98.74±1.01	**98.87±1.34**	96.42±3.61	98.58±1.30	91.08±5.11	89.98±2.91	96.48±2.68
15	76.15±6.01	62.51±5.97	65.26±11.38	70.39±6.43	82.49±12.79	81.01±7.21	84.33±18.44	78.14±7.01	82.41±7.54	66.99±14.98	**87.04±6.96**
16	89.50±10.02	98.81±0.96	98.74±0.82	**99.72±0.36**	97.91±1.72	97.00±3.22	98.53±0.64	97.62±1.24	96.29±3.77	96.07±2.75	99.27±1.00
OA (%)	90.96±1.13	86.56±1.44	88.82±2.66	91.08±1.84	91.80±0.95	91.72±1.86	94.24±1.77	92.10±1.64	89.17±0.85	81.98±2.03	**95.12±0.85**
AA (%)	93.00±0.93	90.14±1.69	94.74±0.69	94.36±1.31	96.28±0.66	95.82±0.59	96.69±1.01	96.16±0.75	92.37±0.70	90.05±0.61	**97.27±0.53**
$\kappa\times100$	89.97±1.23	85.06±1.59	87.59±2.92	90.09±2.04	90.89±1.06	90.80±2.06	93.58±1.99	91.22±1.82	87.99±0.93	80.05±2.17	**94.57±0.95**

the highest averaged OA among the six classes. As shown in Fig. 10, DM-MRN produces a clearer classification map compared with other methods. However, the relatively low spatial resolution and few training samples pose great challenges to different classification methods. Specifically, all the methods mistakenly identify *Sea* as *Salt filter pond*, *Salt evaporation pond*, *Pit pond*, or *Fishpond* due to cloud cover in this area. In contrast, DM-MRN can partially alleviate this

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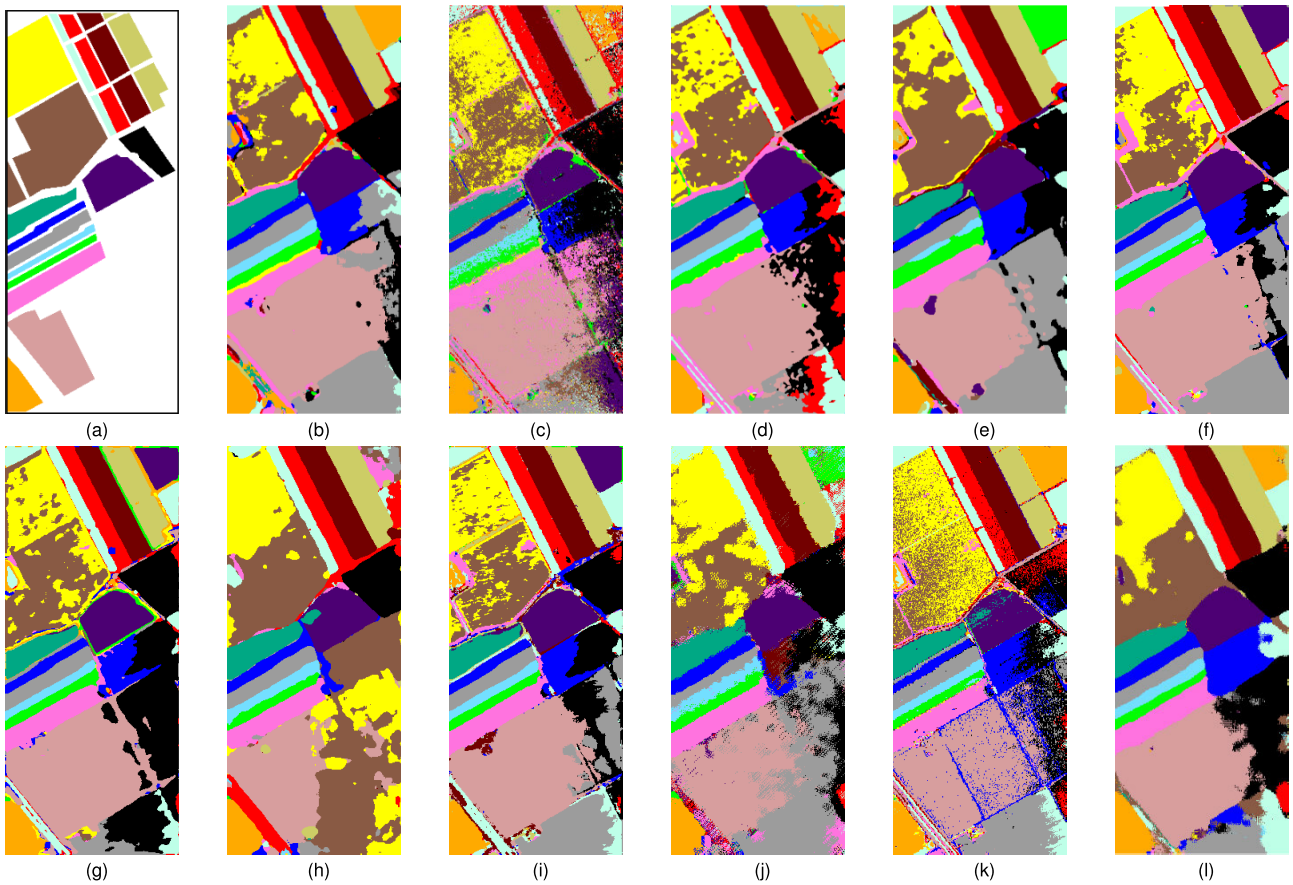


Fig. 9. Classification maps obtained by different methods for Salinas. (a) Ground-truth, (b) 2-D-CNN (90.96%), (c) 3-D-CNN (86.56%), (d) SSRN (88.82%), (e) DPN-HRA (91.08%), (f) S-DMM (91.80%), (g) CMFSL (91.72%), (h) S3Net (94.24%), (i) DCFSL (92.10%), (j) SSFTT (89.17%), (k) miniGCN (81.98%), and (l) DM-MRN (95.12%).

TABLE IX

CLASSIFICATION ACCURACIES OBTAINED BY DIFFERENT METHODS FOR YELLOW RIVER DELTA (USING TEN LABELED SAMPLES PER CLASS FOR TRAINING)

Class	2D-CNN [20]	3D-CNN [20]	SSRN [21]	DPN-HRA [43]	S-DMM [57]	CMFSL [58]	S3Net [41]	SSFTT [27]	miniGCN [33]	DM-MRN (Ours)
1	74.95±19.98	68.99±10.64	63.56±21.82	**82.10±6.77**	72.83±12.18	67.67±8.26	78.37±6.32	66.91±10.05	44.53±9.25	75.30±13.42
2	69.66±19.71	75.10±18.50	**97.26±4.13**	78.86±12.06	96.16±4.47	95.76±3.59	90.85±6.95	88.41±5.88	94.69±3.65	93.90±7.15
3	89.04±5.88	93.26±7.25	91.57±18.91	86.77±7.84	97.47±5.08	98.02±5.81	**98.27±5.19**	95.90±5.25	97.00±5.20	96.20±5.57
4	84.33±12.09	86.72±8.03	87.26±15.29	86.63±12.63	96.03±5.02	96.21±5.49	**98.83±3.52**	97.10±4.75	97.00±5.21	96.07±4.35
5	93.14±6.21	92.54±4.23	88.27±14.87	94.56±8.85	97.54±3.01	96.93±3.08	93.23±4.73	96.07±2.40	94.92±1.76	**97.62±3.31**
6	75.41±7.12	61.47±19.49	72.19±15.75	76.50±6.66	**93.33±5.94**	88.21±5.54	88.55±5.06	86.05±6.79	85.30±7.84	89.49±4.46
7	77.99±21.35	92.52±14.09	99.63±1.10	86.61±12.55	99.86±0.31	99.97±0.10	**100.00±0.00**	96.48±2.73	99.36±0.10	99.83±0.41
8	95.06±2.35	96.43±2.62	96.85±2.84	96.99±2.37	98.51±1.57	97.50±4.26	95.87±6.40	96.15±2.95	**99.42±0.85**	92.93±5.85
9	99.42±0.96	98.10±3.63	99.74±0.33	99.69±0.55	**100.00±0.0**	99.03±0.71	100.00±0.00	97.03±1.99	100.00±0.00	99.67±0.66
10	99.56±0.43	98.76±0.79	99.29±0.51	**99.78±0.20**	82.81±8.24	86.04±4.49	82.57±8.16	86.92±4.73	87.28±6.80	85.25±10.38
11	78.12±22.87	70.08±23.36	47.42±34.19	68.14±30.54	100.00±0.0	94.29±17.14	97.14±8.57	94.29±17.14	70.00±19.64	**100.00±0.00**
12	67.46±20.69	13.52±6.72	34.15±32.94	34.71±22.22	**94.39±5.54**	89.82±8.41	92.46±6.18	82.25±6.58	42.11±11.66	92.63±6.74
13	91.79±6.86	94.09±3.78	90.41±9.51	**96.23±4.04**	92.52±5.99	93.32±6.69	88.29±10.97	85.47±11.62	93.21±6.34	90.85±7.01
14	82.86±12.58	69.21±21.42	65.52±30.10	78.45±24.04	86.10±3.67	86.07±7.18	91.67±5.60	87.71±4.61	70.27±10.23	**91.87±5.80**
15	64.69±7.76	84.82±8.38	66.15±24.69	58.35±5.91	99.31±2.08	97.92±2.36	99.80±0.59	95.84±4.83	98.61±2.98	**100.00±0.00**
16	59.39±22.31	61.18±12.29	78.15±14.68	64.02±16.78	92.46±7.15	96.40±7.32	**97.72±4.30**	89.63±9.17	71.23±5.46	96.14±7.11
17	41.83±32.68	51.82±26.44	67.77±34.51	59.44±22.83	93.14±5.63	90.17±7.50	94.66±6.16	88.35±7.76	78.56±12.19	**97.03±2.82**
18	93.42±7.82	95.62±3.62	91.60±15.11	**96.07±4.64**	94.46±6.01	90.64±4.93	95.15±2.61	83.28±13.80	83.71±8.98	94.64±3.66
19	78.76±17.10	75.33±16.30	51.06±18.76	75.14±11.57	**99.74±0.78**	99.61±0.83	92.08±12.09	92.46±5.18	91.69±10.85	98.83±2.69
20	92.01±4.97	87.00±5.90	87.18±4.34	**92.29±3.93**	86.12±9.51	90.66±4.45	92.15±6.05	80.47±6.42	82.99±13.86	91.67±4.29
21	77.93±6.40	75.25±12.26	71.35±26.99	83.68±4.34	**84.94±4.57**	77.02±8.41	77.86±7.87	75.57±9.53	54.16±12.91	81.34±7.60
22	60.13±13.80	50.42±15.37	45.92±22.23	60.00±10.79	96.72±3.59	94.92±2.48	99.34±1.97	88.85±11.01	84.59±10.03	**99.67±0.98**
23	92.79±6.13	94.32±6.81	**99.89±0.34**	96.15±5.27	99.03±0.32	99.03±0.32	97.63±2.14	95.36±4.72	93.98±2.64	98.71±1.43
OA (%)	86.13±4.22	83.07±5.90	81.89±9.48	88.55±4.46	87.88±3.96	88.96±2.10	87.86±4.05	87.42±3.11	85.98±3.77	**89.26±5.10**
AA (%)	79.99±3.99	77.68±3.70	77.92±6.01	80.49±3.66	93.63±0.78	92.40±1.01	93.15±1.07	88.98±2.43	83.24±0.86	**93.90±1.14**
$\kappa\times100$	82.42±4.92	78.85±6.60	77.55±10.38	85.36±5.34	84.65±4.60	85.81±2.47	84.63±4.83	83.77±3.83	81.92±4.57	**86.40±6.03**

problem, which may be due to the powerful ability of feature learning.

4) Berlin: According to the results listed in Table X, the accuracies of DM-MRN, DPN-HRA, and CMFSL are obviously higher than other methods. Specifically, DM-MRN achieves the highest averaged OA with 61.64%, which increases from 1.04% to 12.29% compared with other methods. It is worth noting that the OA obtained by all methods

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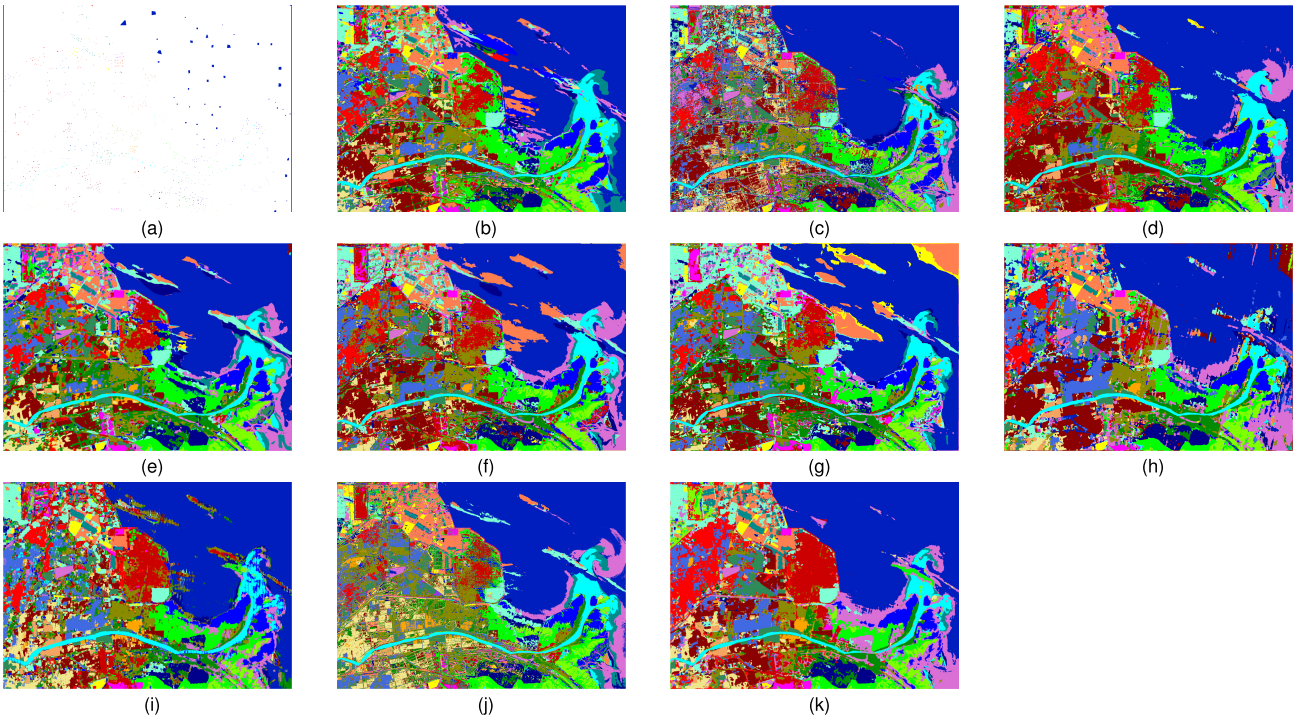


Fig. 10. Classification maps obtained by different methods for Yellow River Delta. (a) Ground-truth, (b) 2-D-CNN (86.13%), (c) 3-D-CNN (83.07%), (d) SSRN (81.89%), (e) DPN-HRA (88.55%), (f) S-DMM (87.88%), (g) CMFSL (88.96%), (h) S3Net (87.86%), (i) SSFTT (87.42%), (j) miniGCN (85.98%), and (k) DM-MRN (89.26%).

TABLE X
CLASSIFICATION ACCURACIES OBTAINED BY DIFFERENT METHODS FOR BERLIN (USING TEN LABELED SAMPLES PER CLASS FOR TRAINING)

Class	2D-CNN [20]	3D-CNN [20]	SSRN [21]	DPN-HRA [43]	S-DMM [57]	CMFSL [58]	S3Net [41]	DCFSL [59]	SSFTT [27]	miniGCN [33]	DM-MRN (Ours)
1	71.57±7.77	62.05±8.72	72.27±7.08	**84.95±7.51**	76.99±9.97	77.04±3.75	79.50±11.33	72.61±7.07	75.22±6.96	74.18±14.79	73.67±7.29
2	**93.54±1.81**	87.81±1.71	91.06±2.76	91.57±2.71	45.54±6.85	53.84±2.98	42.86±11.14	53.83±10.30	52.30±9.39	38.62±9.38	56.45±7.83
3	25.52±7.96	23.96±8.32	28.10±13.56	31.79±13.32	52.23±7.10	51.78±7.67	**62.23±7.84**	43.06±11.36	47.13±8.84	37.67±11.95	54.50±7.75
4	79.90±5.24	79.95±5.19	78.82±6.26	**85.49±7.82**	69.40±9.52	73.82±6.74	68.15±8.54	62.79±9.52	67.32±7.17	67.40±8.06	71.14±8.08
5	55.76±10.79	56.31±11.26	59.68±13.39	64.40±7.31	**90.94±4.19**	87.86±4.80	86.80±8.49	85.63±8.37	88.58±7.35	75.32±14.73	83.19±11.22
6	17.59±4.41	13.69±3.07	16.29±5.83	18.48±6.07	78.06±5.10	68.83±7.12	74.08±11.10	59.79±6.67	68.80±9.51	69.73±10.92	**78.13±7.25**
7	21.03±6.62	15.29±3.67	18.07±4.67	22.96±7.52	44.25±7.68	40.19±9.63	31.30±13.35	37.33±6.49	39.00±4.62	45.04±9.39	**45.21±11.92**
8	30.77±7.77	12.84±3.75	47.93±19.89	20.54±8.57	**87.48±2.00**	82.26±5.94	82.35±8.37	75.15±11.35	57.87±11.75	71.80±7.64	80.23±6.88
OA (%)	57.45±7.97	49.35±1.92	53.72±6.37	60.60±6.79	55.75±3.98	60.43±2.37	53.72±7.38	57.53±6.93	57.91±6.36	49.54±5.67	**61.64±4.87**
AA (%)	49.46±3.47	43.99±1.63	51.53±3.55	52.52±2.56	**68.11±2.05**	66.95±1.63	65.91±2.29	61.27±5.91	62.03±2.98	59.97±4.01	67.81±2.39
$\kappa\times100$	45.75±7.02	36.58±1.51	41.56±5.98	47.72±6.75	43.93±3.76	47.95±2.48	42.31±6.15	43.93±7.03	45.19±6.26	37.70±5.01	**49.14±4.77**

on this dataset is not superior. As can be seen from Fig. 11, *Residential Area* is always wrongly identified as *Allotment*, *Commercial Area*, and *Water*. We consider this is caused by the low resolution of this image and the few labeled samples (only 0.02% labeled samples) used for training. But in contrast, DM-MRN can generate a clearer classification map.

5) Houston2018: According to the results listed in Table XI, the accuracies of DM-MRN, S-DMM, and CMFSL are obviously higher than other methods. Specifically, DM-MRN achieves the highest averaged OA with 55.68%, which increases from 4.34% to 19.21% compared with other methods. Besides, the AA and κ of DM-MRN are also obviously improved. In terms of class-specific accuracy, DM-MRN has the highest averaged OA among the nine classes. As shown in Fig. 12, the classification map of DM-MRN is obviously less noisy and smoother. It is worth noting that the classification map of DM-MRN is the smoothest in *Unpaved parking lots*, *Trains*, and *Railways*, which is consistent with that reported in Table XI.

IV. DISCUSSION

A. Different Combinations of Two Metrics

To demonstrate the influence of different combinations of two metrics, we design four multistage relation networks whose relation modules are composed of "I2C + I2C + I2C," "I2C + I2I + I2I," "I2I + I2I + I2I," and "I2C + I2C + I2I (Ours)." As shown in Table XII, the averaged OA obtained by the multistage relation network composed of two metrics is higher than that composed of one metric, which is in accordance with our former observation. It's interesting to note that the averaged OA of "I2C + I2C + I2I (Ours)" is always higher than "I2C + I2I + I2I." We consider that this is due to the fact that the I2C block computes similarity by second-order features, which can summarize more complex and abstract information for classification.

B. Feature Separability Analysis

To verify the feature separability of the proposed method, we transform the feature maps into 2-D space by using

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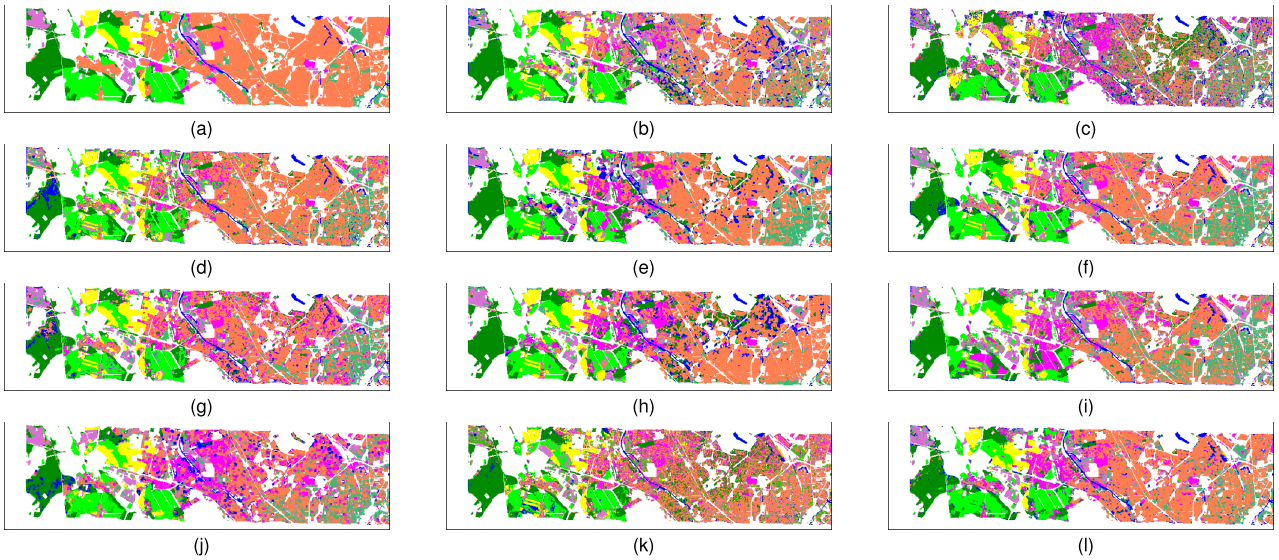


Fig. 11. Classification maps obtained by different methods for Berlin. (a) Ground-truth, (b) 2-D-CNN (57.45%), (c) 3-D-CNN (49.35%), (d) SSRN (53.72%), (e) DPN-HRA (60.60%), (f) S-DMM (55.75%), (g) CMFSL (60.43%), (h) S3Net (53.72%), (i) DCFSL (57.53%), (j) SSFTT (57.91%), (k) miniGCN (49.54%), and (l) DM-MRN (61.64%).

TABLE XI

CLASSIFICATION ACCURACIES OBTAINED BY DIFFERENT METHODS FOR HOUSTON2018 (USING TEN LABELED SAMPLES PER CLASS FOR TRAINING)

Class	2D-CNN [20]	3D-CNN [20]	SSRN [21]	DPN-HRA [43]	S-DMM [57]	CMFSL [58]	S3Net [41]	SSFTT [27]	miniGCN [33]	DM-MRN (Ours)
1	37.56±6.67	42.04±5.01	58.91±13.01	45.66±11.48	86.62±4.66	89.19±4.93	76.07±20.15	69.77±5.82	**96.12±3.48**	75.29±8.53
2	76.30±5.86	68.89±7.46	**82.42±6.54**	81.18±2.62	76.38±6.14	78.35±5.92	50.47±18.90	66.43±10.85	67.47±8.75	72.18±5.99
3	37.34±14.35	71.50±22.72	73.59±17.43	39.57±30.39	99.94±0.10	97.55±3.72	95.82±8.44	98.96±2.63	99.69±0.34	**99.99±0.04**
4	57.67±11.03	51.36±7.74	78.39±4.14	67.37±12.32	**92.63±2.96**	91.05±5.20	78.19±28.52	87.54±4.65	89.42±5.55	91.70±4.71
5	25.64±7.54	16.50±6.92	25.88±8.50	20.18±7.33	77.26±7.97	66.55±9.81	70.54±20.57	**80.67±4.93**	61.38±12.12	76.82±9.43
6	28.95±8.48	22.31±9.35	28.68±13.94	41.19±19.79	82.94±10.74	83.56±8.62	66.82±34.58	91.34±8.41	91.51±3.15	**95.32±2.54**
7	14.45±9.00	41.72±27.79	47.90±24.94	37.13±30.35	**99.30±1.37**	97.77±2.05	98.75±2.48	95.86±9.53	96.76±1.34	90.27±14.86
8	55.83±7.69	35.81±7.62	47.83±7.41	55.58±6.06	71.72±3.21	69.22±10.50	46.57±19.07	71.11±6.76	56.52±12.32	**75.15±10.01**
9	93.59±2.51	86.72±3.48	92.28±5.93	**94.43±1.53**	41.32±7.20	42.62±6.93	34.80±8.94	39.75±5.02	26.08±11.37	51.84±8.74
10	**37.44±7.32**	27.51±6.41	32.64±7.47	35.78±8.79	30.54±6.71	29.13±4.15	12.11±10.18	27.27±6.08	14.24±7.15	28.02±6.65
11	29.84±7.06	20.34±5.65	**39.38±6.37**	30.16±4.91	35.24±5.20	30.52±6.94	14.89±12.04	28.03±3.66	15.51±5.65	28.60±6.85
12	2.60±1.04	1.17±0.23	2.45±0.32	1.97±0.81	**53.47±8.63**	41.20±4.65	19.56±15.39	42.44±9.18	20.14±7.74	47.33±10.25
13	**54.57±6.56**	38.66±8.22	51.67±6.51	49.74±7.03	35.05±9.72	31.13±4.38	27.15±18.41	42.14±8.38	19.73±9.68	36.65±11.19
14	37.75±10.31	27.12±2.93	37.76±10.61	41.77±10.67	82.21±10.41	78.70±6.19	62.00±27.69	84.59±10.14	78.49±4.85	**92.74±11.12**
15	55.99±17.05	33.53±8.64	50.42±16.57	50.30±11.62	96.09±2.20	93.26±2.71	78.47±28.20	88.49±4.06	91.31±10.27	**96.69±2.00**
16	41.29±8.81	37.84±10.53	41.30±11.21	46.31±16.46	**85.02±3.13**	75.31±5.96	35.62±23.81	83.49±7.46	62.60±10.19	78.97±8.67
17	2.42±1.24	3.39±1.35	13.15±13.65	9.32±4.36	100.00±0.0	98.92±1.86	89.28±29.39	99.14±2.36	96.76±6.18	**100.00±0.0**
18	20.00±7.69	13.49±3.28	16.13±7.34	16.63±6.04	87.19±5.95	74.47±12.80	49.70±30.66	79.14±5.67	56.81±13.97	**89.65±3.11**
19	34.72±15.40	17.37±7.04	32.55±10.08	27.70±18.61	89.91±3.59	77.12±10.28	53.46±26.95	85.66±5.59	78.53±7.26	**93.84±5.57**
20	28.01±13.37	30.40±7.57	45.68±13.57	43.69±19.47	80.58±10.00	77.96±5.87	72.76±26.76	93.56±5.20	88.59±7.32	**96.02±4.18**
OA (%)	48.04±4.74	36.47±2.18	49.27±5.70	47.14±6.22	51.34±3.04	50.24±3.21	37.38±9.03	49.37±2.73	37.46±4.68	**55.68±4.32**
AA (%)	38.60±1.96	34.38±1.94	44.95±3.19	41.78±3.29	75.17±1.29	71.18±1.63	56.65±12.85	72.77±1.29	65.38±1.43	**75.85±1.39**
$\kappa\times100$	40.99±4.00	29.63±1.94	41.91±5.18	40.36±5.17	44.67±2.63	43.21±2.96	30.21±8.66	42.66±2.56	31.72±3.52	**48.50±4.22**

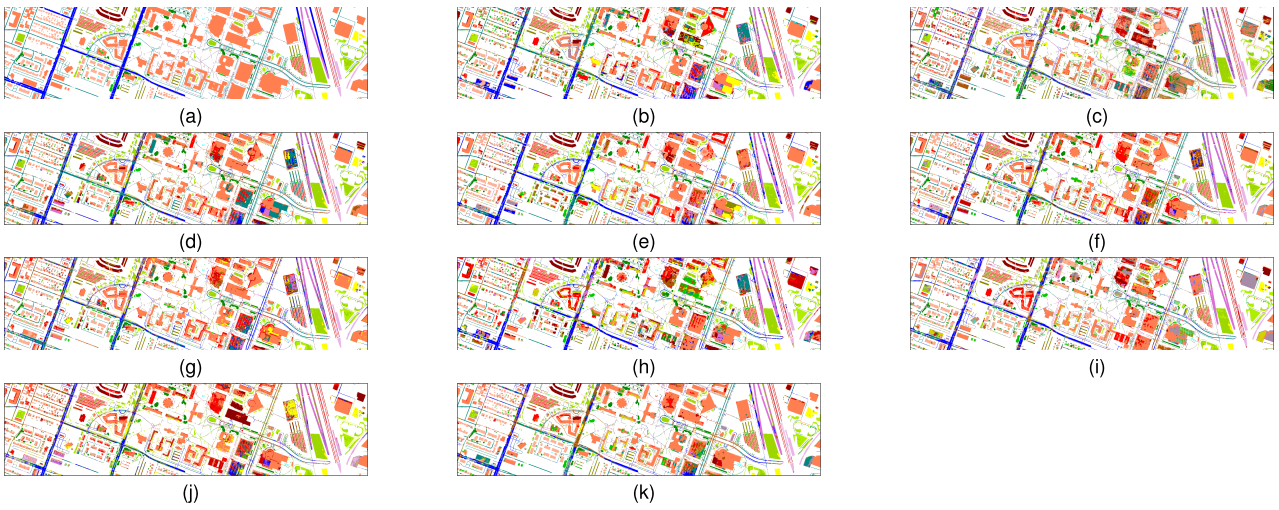


Fig. 12. Classification maps obtained by different methods for Houston2018. (a) Ground-truth, (b) 2-D-CNN (48.04%), (c) 3-D-CNN (36.47%), (d) SSRN (49.27%), (e) DPN-HRA (47.14%), (f) S-DMM (51.34%), (g) CMFSL (50.24%), (h) S3Net (37.38%), (i) SSFTT (49.37%), (j) miniGCN (37.46%), and (k) DM-MRN (55.68%).

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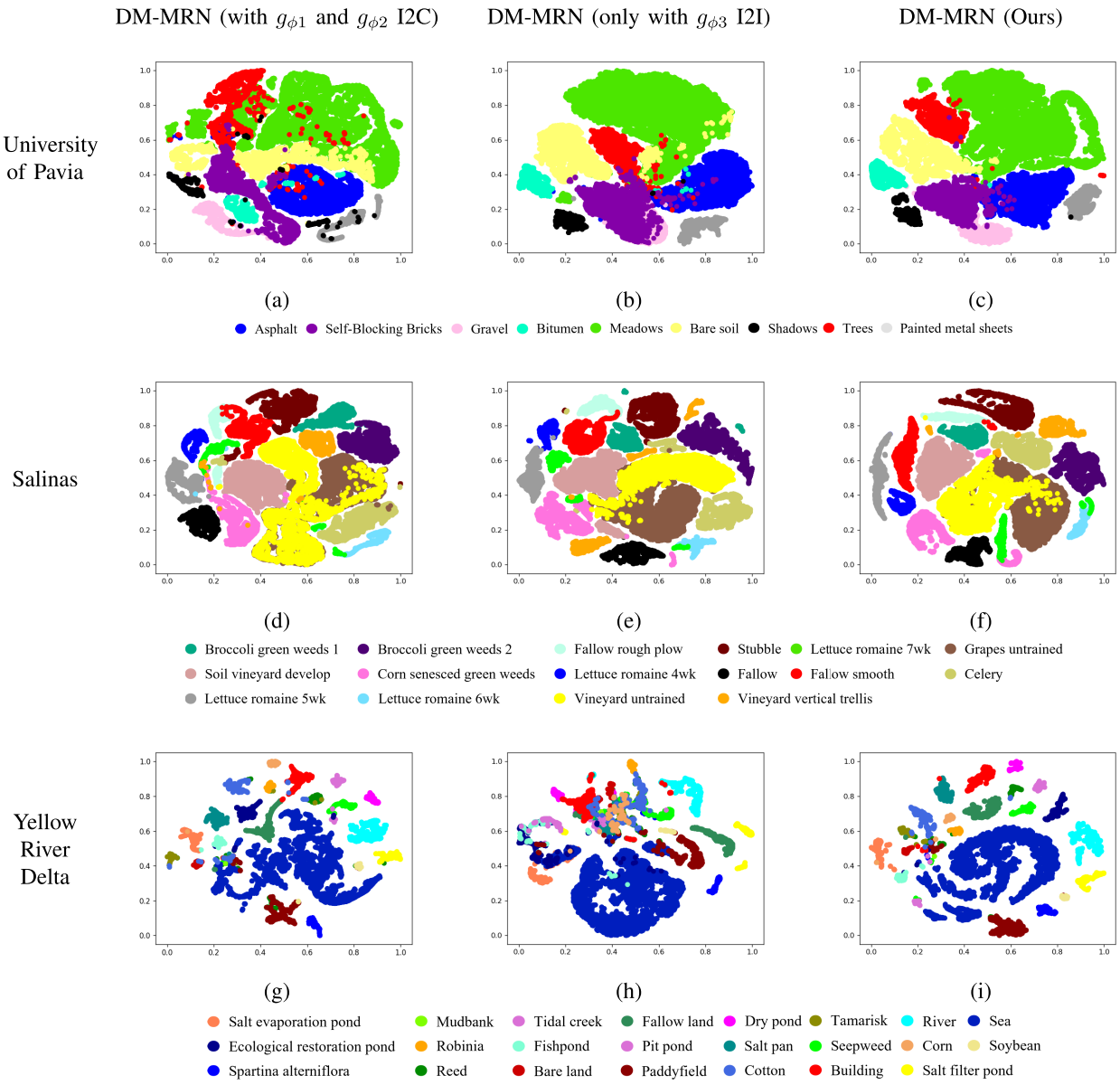


Fig. 13. Visualization of feature separability for different methods. (a) DM-MRN (with $g_{\phi 1}$ and $g_{\phi 2}$ I2C block), (b) DM-MRN (only with $g_{\phi 3}$ I2I block), (c) DM-MRN (Ours) for University of Pavia, (d) DM-MRN (with $g_{\phi 1}$ and $g_{\phi 2}$ I2C block), (e) DM-MRN (only with $g_{\phi 3}$ I2I block), (f) DM-MRN (Ours) for Salinas, (g) DM-MRN (with $g_{\phi 1}$ and $g_{\phi 2}$ I2C block), (h) DM-MRN (only with $g_{\phi 3}$ I2I block), and (i) DM-MRN (Ours) for Yellow River Delta.

TABLE XII
COMPARISON OF DIFFERENT COMBINATIONS OF TWO METRICS

Combination	University of Pavia	Salinas	Yellow River Delta
I2C+I2C+I2C	94.67±1.89	95.31±0.88	93.35±3.69
I2C+I2I+I2I	96.45±1.83	96.31±0.53	94.75±2.27
I2I+I2I+I2I	95.92±2.31	96.21±0.67	95.72±3.01
I2C+I2C+I2I (Ours)	**96.55±2.05**	**96.88±0.85**	**95.83±2.72**

t-distributed stochastic neighbor embedding (t-SNE) [60] and visualize the feature separability, as shown in Fig. 13. For the University of Pavia dataset, DM-MRN (with $g_{\phi 1}$ and $g_{\phi 2}$ I2C block) and the DM-MRN (only with $g_{\phi 3}$ I2I block) are similar, which cannot clearly divide the boundaries between *Gravel*, *Trees*, *Self-Blocking Bricks*, *Asphalt*, and *Meadows*. However, the feature separability of DM-MRN (Ours) is obviously better than the above two methods. For the Salinas dataset, the feature separability of the three methods is similar, which is caused by the regular distribution of land covers. For Yellow River Delta dataset, DM-MRN (only with $g_{\phi 3}$ I2I block) tends to confuse *Building*, *Cotton*, *Robinia*, *Salt pan*, and *Mudbank*, whereas features in DM-MRN (with $g_{\phi 1}$ and $g_{\phi 2}$ I2C block) have larger interclass distance. It's noted that DM-MRN (Ours) has the best feature separability among the three methods. In general, compared with DM-MRN only with I2C or I2I block, DM-MRN with two metrics can significantly improves the class distribution with limited samples.

C. Complexity Analysis

To analyze the complexity of DM-MRN, we compared the running time and parameters size of 2-D-CNN, 3-D-CNN, SSRN, DPN-HRA, S-DMM, CMFSL, S3Net, and DCFSL under ten labeled samples in three datasets. As can be seen from Table XIII, S-DMM has the fewest parameters compared

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TABLE XIII
RUNNING TIME (S) AND PARAMETER SIZE OF DIFFERENT METHODS (TEN LABELED SAMPLES PER CLASS)

Data sets	Metrics	2D-CNN [20]	3D-CNN [20]	SSRN [21]	DPN-HRA [43]	S-DMM [57]	CMFSL [58]	S3Net [41]	DCFSL [59]	DM-MRN (Ours)
University of Pavia	Train	177.26	150.88	70.86	81.09	1509.71	690.48	36.45	2042.43	82.88
	Test	109.37	39.70	155.32	116.19	66.67	2.58	38.17	6.32	50.52
	Parameter size	105641	1067945	216537	1161637	33921	614006	91517	4259294	181782
Salinas	Train	193.73	258.27	192.06	64.87	1633.14	1075.04	141.23	5344.98	159.72
	Test	113.83	51.53	422.24	131.44	94.33	3.16	148.09	81.52	104.87
	Parameter size	113712	1121904	370312	1161637	40385	614006	155008	4270521	183062
Yellow River Delta	Train	180.25	384.01	87.26	92.06	27396.33	2221.23	242.81	-	773.55
	Test	20.24	8.99	27.69	24.20	16.33	5.46	23.25	-	25.06
	Parameter size	121783	1084215	241463	1161637	34945	614006	198019	-	183062

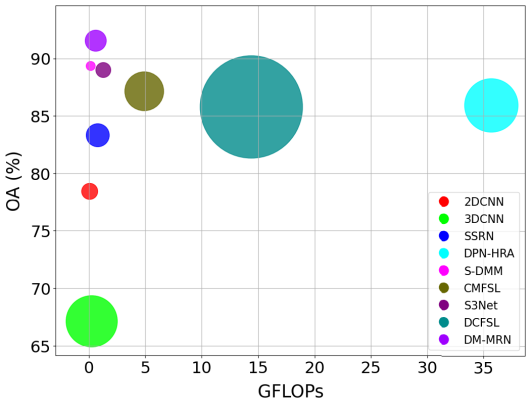


Fig. 14. Comparison of OA, FLOPs, and parameter size of each model in the University of Pavia dataset (ten labeled samples per class).

with other methods. This is because the structure of S-DMM is very concise, and only 2-D convolution is employed in its construction. However, due to the 50 000 iterations during the training period, S-DMM is a time-consuming method among these methods. DM-MRN has more parameters than S-DMM, which is due to the fact that two improved metric blocks are used in the relation module. Even so, the parameters of DM-MRN are still smaller than those of other methods, and it is suitable for small sample classification.

In addition, we also collect the FLOPs of each model when it is trained in the University of Pavia dataset with ten labeled samples per class. Fig. 14 shows our obtained results. The abscissa represents GFLOPs (1 GFlops $= 10^9$ Flops) and the ordinate represents OA. Each circle in the table represents a model whose radius is proportional to the model's parameter size. It can be seen that S-DMM, S3Net, and DM-MRN can strike a good balance among OA, parameters, and FLOPs, which have the characteristics of lightweight networks.

V. CONCLUSION

In this article, we propose a novel DM-MRN for HSI classification. The innovative contributions of DM-MRN include three aspects. First, a novel I2C block is designed to condense class-level features through second-order features. Second, a novel I2I block is conceived to learn representative feature relations from the channel and spatial dimensions. Third, a multistage network (DM-MRN) with two metrics is constructed. Experiments on five HSI datasets demonstrate that our proposed method not only has excellent generalization performance but also has the highest classification performance compared with other traditional and advanced DL-based HSI classification methods. In addition, we also verify the effectiveness of the proposed I2C block, I2I block, dual-metric, and multistage structure. At the same time, we compare the multistage structure composed of different metric combinations, analyze the feature separability, and calculate the complexity of each model.

Although we have obtained encouraging experimental results, we need to further improve the proposed model from two research lines. On the one hand, the current method only trains and predicts one dataset. In the future, we will improve the model using episode training based on multiple datasets, and then transfer the model to a new dataset after fine-tuning. On the other hand, a multistage relation network contains three stages. We will try to discuss the influence of different number of stages on classification performance.

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